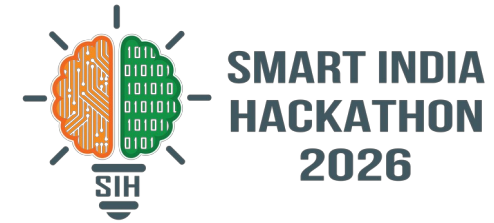
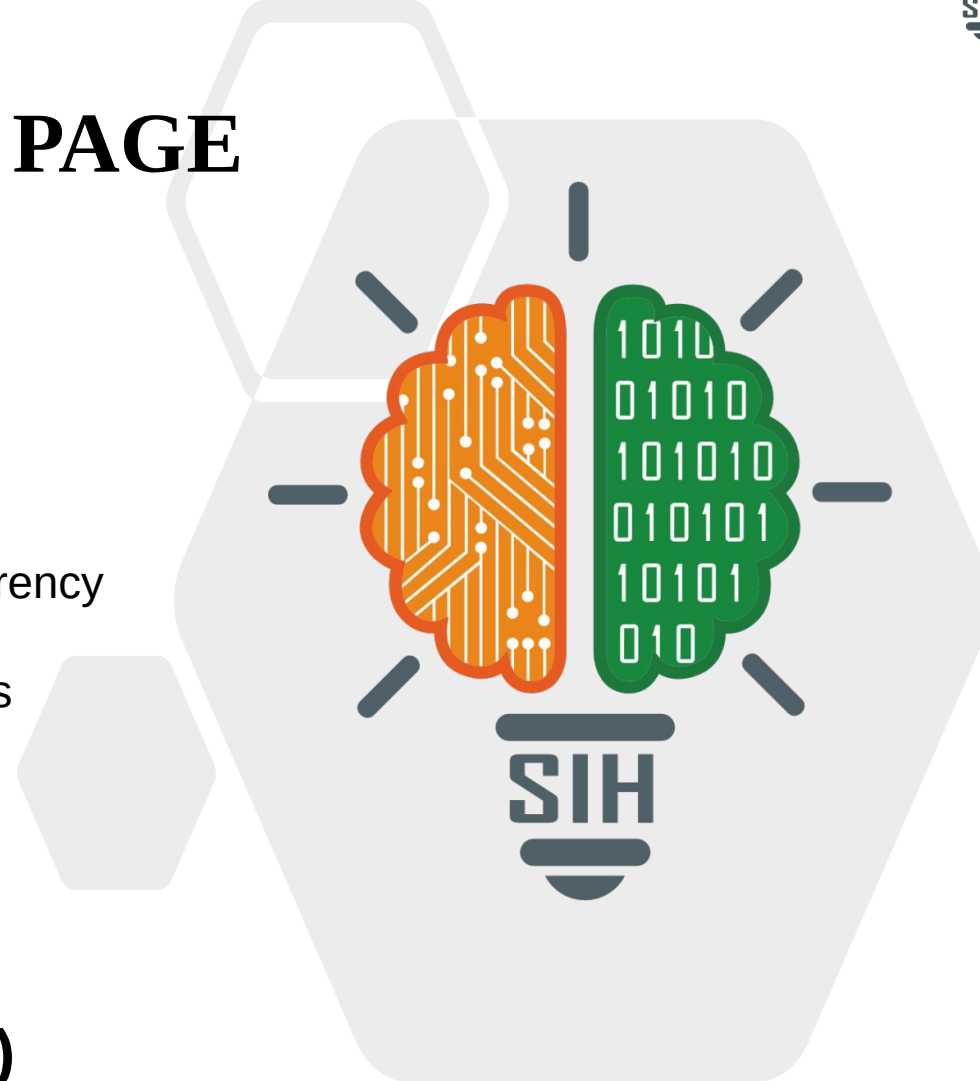


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26183
- **Problem Statement Title-**
Real-Time Identification of Fraud-Linked Cryptocurrency
Exchanges from Victim-Reported Suspect Wallet
Addresses through Automated Blockchain Analytics
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-** 156766
- **Team Name (Registered on portal)**
COLD COFFEE

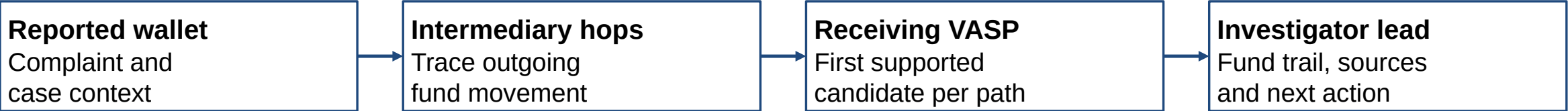




AEGIS FORENSICS

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation of the proposed solution



- How it addresses the problem

Manual block-explorer tracing requires investigators to follow transactions individually, check destination labels and revisit addresses for new activity. AEGIS will automate these steps and prioritize receiving-VASP candidates for investigator review.

Each lead will include the transaction path, attribution source, uncertainty and recommended next action, supporting faster preservation requests and VASP coordination.

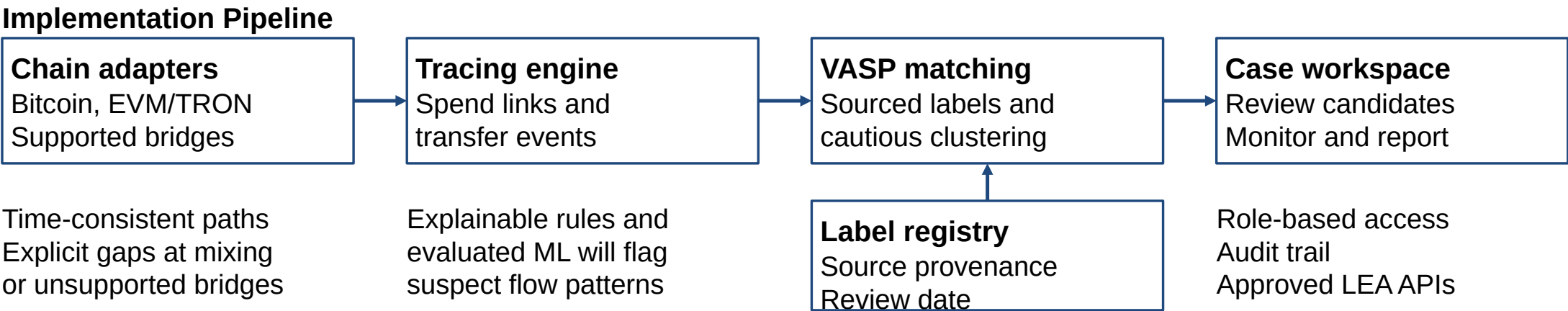
- Innovation and uniqueness of the solution

Continuous monitoring paired with explainable attribution: new transactions or revised wallet labels will update the investigation and trigger relevant alerts.

Every VASP lead will show why it was identified, which links are observed or inferred, and where tracing remains unresolved. Versioned findings will preserve the basis of earlier conclusions.

- Technologies to be used (e.g. programming languages, frameworks, hardware)
 React investigator dashboard; FastAPI services; queued tracing and monitoring workers; chain-specific data adapters; incremental transaction indexing; persistent case storage; and a versioned VASP-label registry.

- Methodology and process for implementation (Flow Charts/Images/ working prototype)



Findings will distinguish observed transactions, inferred links and unresolved attribution.

- Analysis of the feasibility of the idea



Label supply: reviewed public disclosures and authorized VASP data, with source history.

- Potential challenges and risks

Uncertain labels and flows

Provider limits and large histories

Sensitive cases and restricted access

- Strategies for overcoming these challenges

Version label sources, review clusters and show unresolved paths.

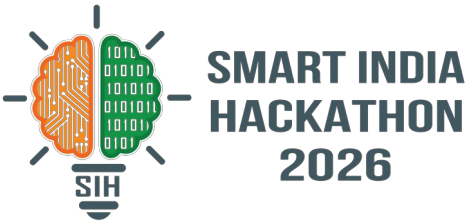
Index incrementally, cache queries and bound graph traversal.

Role-based permissions, audit logs and approved NCRP/SAHYOG APIs.

Validation: held-out labelled paths and hard negatives. Measure precision, coverage and alert latency.



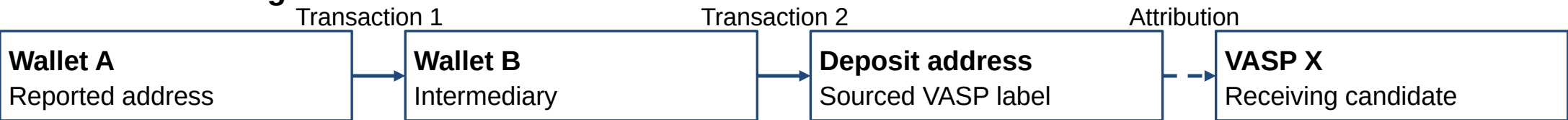
IMPACT AND BENEFITS



- Potential impact on the target audience

Investigators will receive a prioritized receiving-VASP lead with a reviewable fund trail.

Illustrative investigation



Finding Two-hop path, transaction IDs, amounts, timestamps and label source.

Next action Investigator reviews the lead and prepares a VASP preservation request.

Fictional illustration. Transaction paths alone do not prove exact fund continuity or wrongdoing.

- Benefits of the solution (social, economic, environmental, etc.)

Social

Help investigators act sooner on victim reports and identify potential intervention points. Make complex fund trails understandable without requiring specialist blockchain expertise.

Economic

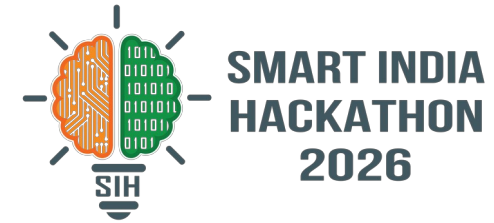
Reduce repeated transaction lookups and manual report preparation. Prioritize actionable leads so investigators can focus limited time on review and response.

Operational

Standardize investigation reports and VASP coordination. Preserve transaction references, label sources and analysis history so findings can be reviewed and handed between teams.



RESEARCH AND REFERENCES



- Details / Links of the reference and research work

Address clustering	Meiklejohn et al. (2013), A Fistful of Bitcoins: Characterizing Payments Among Men with No Names. ACM IMC paper
ML-assisted analysis	Weber et al. (2019), Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics. arXiv:1908.02591
Blockchain data	Blockstream Esplora: address and transaction API documentation. Official API documentation
Attribution sources	Binance, Our Commitment to Transparency (2022): published wallet disclosures. Historical disclosure; not a deposit-account registry
AML context	FATF (2021), Updated Guidance for a Risk-Based Approach to Virtual Assets and VASPs. FATF guidance
Evidence handling	NIST SP 800-86 (2006), Guide to Integrating Forensic Techniques into Incident Response. NIST guidance; not a guarantee of legal admissibility